

# Adhithya Anandram

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## Professional Summary

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Mechanical Engineer with expertise in precision mechanical design, CAD modeling, and system optimization, with hands-on experience in mechatronics and 3D printing. Skilled at executing full product development cycle from concept to manufacturing.

## Technical Skills

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Product design, SolidWorks (CSWP), DFM, GD&T, PDM, MATLAB, ANSYS, Statistical Analysis, LabVIEW, tolerance stack-up analysis, RCA, DOE, Python, FEA, AutoCAD, ASME Y 14.5, Material Selection, OSHA, Six Sigma, Rheology, FTIR, SEM

## Experience

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**R&D Mechanical Engineer**, Brinter – Winston Salem, NC Oct 2025 – Present

- Designed and validated a resorbable rotator cuff repair implant by using custom G-code and optimized FDM process parameters for PLLA core fabrication, followed by extrusion-based collagen printing to encapsulate the core — achieving a 71% improvement in tensile strength.
- Authored manufacturing, testing, and quality control SOPs in alignment with ISO 13485 QMS requirements; executed CAPA protocols to address process deviations identified through mechanical testing, SEM, FTIR, and rheological characterization, sustaining product consistency across production batches.
- Conducted implant manufacturing within an ISO 5-classified xVivo system, maintaining environmental controls and traceability of individual process conditions to support pilot animal study.
- Engineered an automated CAD-to-topology-optimization pipeline using Python code integrating Onshape and nTop that reduced prototyping cycle time by 3×, and extended the design framework to a micro-porous 3D breast scaffold made of TPC for post-mastectomy adipose tissue regeneration.

**Visiting Research Assistant**, Digital Manufacturing & Polymeric Materials Lab – Raleigh, NC June 2025 – Dec 2025

- Developed MATLAB-based grayscale mask generation for Gradient Light VP-SL, achieving optimized exposure control that eliminated porosity and improved surface quality by 70%.
- Assembled and programmed precision linear stage for hybrid system combining extrusion, DLP, and sintering modules, enhancing platform versatility.
- Mentored high school student on layer adhesion optimization; achieved 30% reduction in peel forces using silicone-infused PDMS.
- Developed a FEA model in ANSYS to simulate interface debonding between a hyperelastic PDMS substrate and hydrogel for DLP 3D printing, using distance-based debonding to capture separation behavior.

**Graduate Research Assistant**, Digital Manufacturing & Polymeric Materials Lab – Raleigh, NC Aug 2023 – May 2025

- Designed and optimized lab-scale DLP bioprinting workflows, applying DOE and SPC to reduce print variability by 15%.
- Conducted process development experiments with bio-based resins and photoinitiators; prepared detailed documentation.
- Built modular curing stations and developed process control protocols for resin curing and part quality validation.

**Customer application and support Intern**, Rapid Shape – Raleigh, NC Aug 2024 – Dec 2024

- Conducted root cause analysis for recurring failures in Straumann 3D printers, reducing downtime by 20%.
- Documented calibration and process control workflows in compliance with GMP and ISO standards.
- 3D printed 50+ dental models and surgical guides using beige resin on Rapid Shape DLP systems.

**Design Engineer Intern**, Maruti Suzuki – Pondicherry, India Sep 2021 – Mar 2022

- Optimized steering and brake pad assemblies via DFM/DFA, boosting assembly efficiency by 20%.
- Conducted stress analysis and material selection, enhancing component life by 15%.

## Projects

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### In-situ flaw detection for Electron Beam metal additive process

- Developed a Python program with 80% accuracy in detecting pores in EB-PBF using Near Infrared, Back-Scatter Electron imaging; validated with Computed Tomography scans for varying build parameters.
- Performed materials characterization (hardness, density) for the parts printed using M290 Concept laser.

### Digital Twin for vehicle steering and suspension system

- Modeled Formula-car steering and suspension in SolidWorks; ran stress and modal analyses in ANSYS to avoid resonance.

- Performed DOE by varying suspension and steering parameters and developed reduced order models to simulate the dynamic system response under various load conditions, integrating sensor data and real-time visualization for performance optimization.

## Education

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**MS in Mechanical Engineering**, NC State University

Jan 2023 – May 2025

- **Thesis:** "Photopolymerization of Epoxies using Anthracene Photosensitizer for DLP 3D printing" - under Dr. Henry Ware

**BS in Mechanical Engineering**, Pondicherry University

Sept 2017 – Aug 2021